

Team 2 - Billy's Amazing Team

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Introduction (2a):

Planning

For almost all the changes made to the original group's deliverables, we discussed them beforehand as a team in our meetings. We kept track of what the changes were to be in the meeting minutes, assigning a section to an individual. This allowed us to appropriately plan changes and to ensure we were all happy with the expected outcome.

Making

When making changes, we used the Google Docs change tracking feature for deliverable changes, and for the implementation we used GitHub's push tracking feature. This meant two things, the first being that every change was visible to the whole team, ensuring that all group members could keep an eye on changes as they were happening. The second was that it allowed easy undoing of any changes that the group were not in favour of, meaning that changes could be made without the worry of permanent ramifications.

Keeping track of

We employed several strategies to keep track of the changes to the deliverables. In our meetings and meeting minutes, we outlined all changes we planned to or had made, so we were able to use these as a log. We also updated this change report frequently to ensure any changes were reflected in each section. For the changes to the code, the GitHub log has been useful for showing changes as well.

Reviewing changes

Due to us making notes on each change, both on the meeting minutes and in this change report, reviewing changes was easy. We knew exactly what had been changed, so we were able to efficiently check each to ensure they were updated appropriately. Since we kept track of who did what change in the meeting minutes, we could take up any issues with the change owner.

Requirements (2b):

Links: [Original Requirements Document](#) and [New Requirements Document](#)

Changes:

We decided to remain as faithful to the original group's requirements as possible, as this would allow us to focus our time on implementing new features rather than refactoring and altering the existing code and documentation. The first few alterations were minor: converting the document from pdf broke the formatting so that has been redone and the system requirements heading has also been removed for being previously unused.

In addition, 2 new user requirements were added to fulfill the new requirements added to the product brief: these being the UR_SCOREBOARD and UR_ACHIEVEMENTS requirements. These have been set to 'Shall' priority as they are required in the brief.

Four new system requirements have then been added to break down these problems into parts. Three are for the score (FR_CALCULATE_SCORE, FR_DISPLAY_SCORES and FR_STORE_SCORE) and for achievements (FR_CHECK_ACHIEVEMENTS).

We also decided to alter two requirements to fit the new features we were planning to implement or to improve user experience, these are described below:

- We altered FR_SATISFACTION_TRACK, the original text read: The system shall display what is affecting the student satisfaction, *when the satisfaction meter is clicked on*. We altered this to instead integrate with the new 'thoughts' based approach to satisfaction bar reasoning (as set out in their original architecture). The new requirement is: The system shall display what is affecting the student satisfaction, *in the form of student thoughts displayed as text on the screen*.
- We altered NFR_MULTIPLE_CONSTRUCTION, the original text stated that there would be a limit on the number of buildings that can be built/upgraded at once, however, we felt that this was an arbitrary limitation that would simply get in the player's way rather than add any challenge or improve their experience. Due to this we altered the text to read: there shall be no limit on the number of buildings the player can build/upgrade at once.

All other Requirements are unaltered as no previous requirements have been removed from the brief, and we believe that the remaining requirements align with both our goals, as well as the brief. There are unlikely to be any further changes to the document as there will be no more client-team meetings.

There were user and functional requirements that we were not able to meet during the timescale of this project (UR_ACCESSABILITY_SETTINGS, UR_KEYBINDS, FR_COLOUR_BLIND_MODE, FR_MOUSE_SENSITIVITY_SLIDER), as we decided to focus on other requirements that we deemed more important, from both our own opinions as well as the data gathered from our user evaluation. Rather than remove these from the requirements section, we decided to keep them and instead discuss why we did not meet these, as it will allow more transparency in regards to our decision making process when it comes to implementation (these are discussed in deliverable 3b).

Architecture (2b):

Links: [Original Architecture Document](#) and [New Architecture Document](#)

Changes:

In terms of architecture, we have decided to make minimal changes. We will follow the original group's plan of using student thoughts and events, along with a dropdown for the current thoughts. We have decided to add the additional requirements of a high score and leaderboard, as well as achievements. These features will be relatively straightforward to implement, considering the current architecture. They will also add a lot of value to the game. The only thing we have changed of the original architecture is making the Events class no longer abstract as seen in the class diagram. This allows us to instantiate events, and means we will not have to manage events across multiple child classes. This makes the code easier to build off of and is a cleaner implementation. Other than these changes, the only other changes that were made were just some small formatting changes to make the document more readable to others.

We also decided to maintain the previous group's level of abstraction within their UML diagrams. Rather than add every class to the diagrams we decided to stick with the main classes that run the game, omitting smaller classes (such as manager classes, libGDX classes etc) to help remove clutter and make our diagrams easier to understand.

Method Selection and Planning (2b):

Links: [Original Planning Document](#) and [New Planning Document](#)

Changes:

This document required quite a few changes, namely due to the second half of the project not being planned almost at all by the original group. Rather than erasing the original planning set out by the original group, we decided to instead build upon it, evolving their plans rather than drawing our own from scratch. Overall the updates needed for this deliverable were mainly the addition of content, and the altering of dates on old content, the specifics are listed below:

- There were a few minor alterations to the first few sections, namely adding more information about drawing the images used in the game, as well as updating the section on our communication methods as we used WhatsApp whereas the original team used Discord. Other than these changes nothing else was changed within the first two sections of the deliverable (ie the software engineering methods and team management sections). This is because the other group by chance followed an almost identical management style and communication system to the one we used, and we found that this worked best for our group during part 1 so we continued to use it during this stage of the module, meaning barely any changes were needed.
- The work breakdown structure was updated to reflect the new requirements added to the brief, as well as the new deliverables required
- A new Gantt chart was created to plan out our timeline for project 2
- Finally, our biggest change was the alteration and addition of new work packages that reflected the updated needs of project two. For example, we added WP6 and WP7, which were created to plan the user evaluations and new deliverables respectively. Some other work packages needed alterations as well to reflect the changes, or to alter the dates of completion. For example WP4 had its end dates altered as the testing needs to be completed for part 2, rather than for part 1, and needs to test the new changes. Another example is the addition of T3.4 in WP3, this new task was created for the additional implementation needs of project 2.

Risk Assessment and Mitigation (2b):

Links: [Original Risks Document](#) and [New Risks Document](#)

Changes:

For this document a vast amount of change was needed. The first and most minor was a few reformatting alterations which increased the readability of the document as well as making it more consistent with the rest of our deliverables.

The second change was rewording parts of the document to again make it more readable and to ensure that the document includes all the information needed. We felt that the original document had wording that did not accurately reflect our decisions regarding risks, and so by altering how the document was written we were able to get it to the level we desired. A major part of this was changing the document to state that we pooled all our risks together as a team, whereas the original group only assigned some members to identify risks.

Another change was re-assigning the risks to our own group members, and we added our method to do this to the deliverable.

We also removed risks, as we found that these risks were unnecessary due to either being unrelated to the project, or not necessarily presenting as a risk (i.e. is entirely preventable by just following the assessment instructions). The removed risks are shown below:

R11	Business	Sensitive Ideas or topics inside the game that disrespects or misrepresents that can lead to backlash	Moderate	Serious	Consulting experts or doing research making sure that no one gets disrespected	Jacob
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R14	Requirements	Exceeding the page limit or deliverables being not up to standard of formatting requirement of assessment could lead to 0 marks	Moderate	Serious	Assign a team member to check all of the deliverables' are meeting the requirement and it is not exceeding the page limit.	Lily
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R10	Business	Game fails to be engaging and makes users not want to play the game	Moderate	Serious	Play testing more and iterating regularly to make sure that the game is appealing to the outside world.	Hassan
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We also felt the need to add risks that were missed by the original team, as well as alter the description of a risk that was worded poorly, and alter the mitigation of another risk to make it a more concrete mitigation rather than theoretical.

Risk	Before	After
R3 (Altered Description)	Not enough communication of the team might lead to delays or conflicting work.	Not enough communication within the team
Originally R12, became R10 (Altered Mitigation)	Having a backup protocol and insuring that each sensitive data is protected from data loss	Save constantly, use cloud-based apps, push to GitHub constantly and work in small chunks at a time
R12 (New Risk)	N/A	N/A